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Investigating the production of all-solid-state battery composite cathodes by numerical simulation of the stressing conditions in a high-intensity mixer

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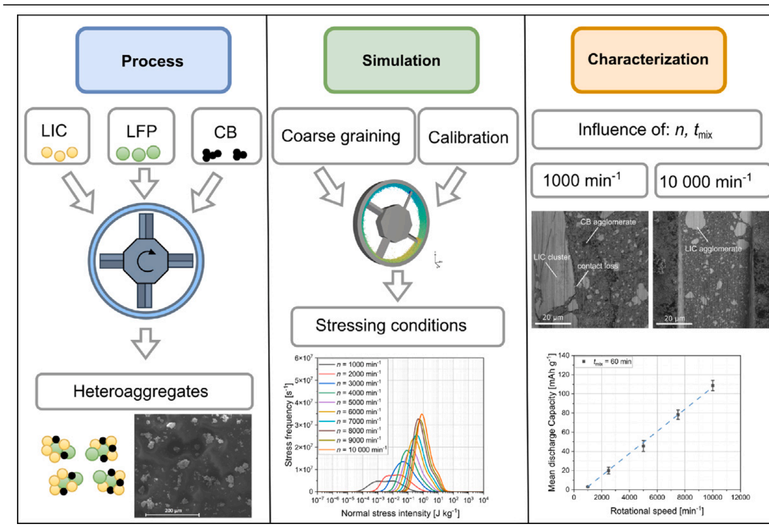
HIGHLIGHTS

- Numerical investigation of the stressing conditions in a high-intensity mixer.
- Application of a new heteroaggregate material system for ASSB.
- Implementation of a force-scaling approach for scaling forces of DEM contact models.
- Calibration of particle behavior via three-step procedure.
- Link between numerical investigations and electrochemical performance of full cells.

GRAPHICAL ABSTRACT



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ABSTRACT

All-solid-state batteries (ASSB) are considered as attractive further development of conventional lithium-ion batteries, since they are expected to deliver higher energy and power densities. So far, ASSB with inorganic solid electrolytes have not been applied in large-scale due to difficulties in processing, insufficient overall ionic conductivities, and high interfacial resistances between active material and solid electrolyte. In this publication, the influence of stressing conditions on the scalable production of ASSB composite cathodes via a high-intensity mixer is investigated. The high-intensity mixer is supposed to circumvent the challenge of high interfacial resistances within an ASSB cathode consisting of submicron-sized active material $LiFePO_4$, carbon black and solid electrolyte Li_3InCl_6. The mixing process is simulated employing the discrete-element-method, from which the stressing conditions acting on the particles are extracted and linked with microstructural observations and electrochemical data. These investigations show that the capacity of the ASSB increases with increasing specific energy input, which is, however, limited by a certain stress intensity determined by the rotational speed of

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the mixer. This examination of the influence of stressing conditions on the production and electrochemical performance of ASSB cathodes lays the foundation for further simulations and experimental investigations in the field of up-scaled ASSB cathode production.

1. Introduction

Lithium-ion batteries (LIB) based on liquid electrolytes will soon reach their development limit [1]. Therefore, new concepts are needed that offer higher energy and power densities [2,3]. All-solid-state batteries (ASSB) are a promising candidate for this purpose, which are also considered safer compared to conventional LIB [4].

In an ASSB, the liquid electrolyte is replaced by a solid electrolyte, which is typically an oxide, a polymer, or a sulfide [5–7]. Oxides such as $Li_{1.4}Al_{0.4}Ge_{1.6}$ (LAGP) or $Li_7La_3Zr_2O_{12}$ (LLZO) are brittle and fragile and therefore difficult to process [8,9]. They form poor contact with the active material of the electrodes, resulting in high interfacial resistances and thus relatively poor electrochemical performance [10–13]. In contrast, oxides show relatively high ionic conductivities in the range of 0.1 $mS\,cm^{-1}$–1 $mS\,cm^{-1}$ at room temperature (RT) [14–17]. For comparison, liquid electrolytes exhibit ionic conductivities of about 1 $mS\,cm^{-1}$–10 $mS\,cm^{-1}$ at RT [1,18]. Polymer electrolytes consisting of polyethylene oxide – lithium bis(trifluoromethanesulfonyl)imide (PEO–LiTFSI) show low ionic conductivities of about 10^{-4} $mS\,cm^{-1}$–10^{-3} $mS\,cm^{-1}$ at RT, which has largely prevented their application [19–22]. However, polymers are easy to process, form good contact with electrodes and are low in price [23,24]. The group of sulfide solid electrolytes like $Li_{10}GeP_2S_{12}$ (LGPS) or Li_6PS_5Cl (LPSCL) exhibit comparably high ionic conductivities of up to 25 $mS\,cm^{-1}$ [25,26]. For this reason and because of their good malleability, sulfides are considered as promising candidates for use in ASSB. Though, a major issue with the use of sulfides is their sensitivity to moisture and oxygen [27,28]. When they come into contact with ambient air, toxic H_2S vapors are formed, thus it is necessary to ensure that sulfides are processed in an argon atmosphere or in a dry room with a low dew point [7,29–32]. In addition, sulfides show instability towards cathode active materials such as $LiNi_xMn_yCO_{1\text{-}x\text{-}y}O_2$ (NMC), $LiCoO_2$ (LCO), and $LiMn_2O_4$ (LMO), which can hinder their use in ASSB [33,34]. To enable the use of sulfides in an ASSB cathode, the active materials NMC/LCO/LMO have to be coated with protective coatings like $LiNbO_3$, or hybrid approaches like a $LiNbO_3 \mid Li_2CO_3$ coating [35–38].

A relatively old class of solid electrolytes that recently attracted new interest are halides such as Li_3YCl_6 (LYC), Li_3YBr_6 (LYB), and Li_3InCl_6 (LIC) [39,40]. In particular, LIC offers comparably high ionic conductivities of up to 2.04 $mS\,cm^{-1}$ at RT and exhibits a higher stability against moisture and cathode active materials such as NMC [40]. The stability towards cathode active materials also allows the use of halides as a protective layer of e.g. NMC, which could allow the use of sulfide solid electrolytes such as LPSCL in combination with a halide coating [41,42].

A major challenge of solid electrolytes such as halides, sulfides, and oxides is their poor contact with the active material [10,43,44]. This usually leads to high charge transfer resistances and eventually to a fast capacity fading and chemo-mechanical degradation of the battery material [45–48]. To improve contact between particles, high intensity mixers can be used to form aggregates or coat particles with other particles, also known as mechanofusion [49–51]. Mechanofusion can be caused by inelastic collisions between particles due to internal and external friction and energy dissipation [52]. Energy dissipation can also lead to aggregate formation due to sinter-like processes [53] and depending on how the energy is dissipated, comminution of larger particles or agglomerates is conceivable.

In the context of batteries, high-intensity mixers have been mainly used for carbon black (CB) desagglomeration which can lead to better cycling results of LIB due to a more homogeneous CB distribution and short-range electronic contacts [54]. However, excessive stress can result in decreased cell performance, as the active material can be coated with CB, resulting in a lack of electronic pathways [55]. In lithium-sulfur batteries coating of sulfur with CB can result in an enhanced cell performance. In these, the volumetric energy density could be increased by about 30% [56] through the formation of core–shell sulfur-CB structures.

Transferred to ASSB, high-intensity mixers could be used to produce aggregates consisting of active material, CB and SE. This may lead to good contact between the particles of the active material and the electrolyte, thus overcoming the problem of high interfacial resistances which often occurs when sulfides, oxides, or halides are used as solid electrolytes [3,57,58].

The mechanism leading to the aggregation of the components, as well as the forces and stress intensities that occur in high-intensity mixers, are not yet fully understood. Since this is particularly difficult to determine at the particle level, simulations play an important role in this context. Simulations of a high-intensity mixer have already been used for studying CB desagglomeration [59]. For this purpose, the discrete-element-method (DEM) was used to determine the frequency and intensity of the stresses and population balances were applied to draw conclusions about CB desagglomeration.

In the present study, a high-intensity mixing process was used to prepare an ASSB cathode heteroaggregate model system consisting of submicron-sized $LiFePO_4$ (LFP), CB and LIC. LIC is intended to act as a highly ion-conducting solid electrolyte phase, which, in contrast to other sulfide solid electrolytes such as LPSCL, should be stable towards the cathode-active material LFP [60]. LFP was chosen because, in contrast to NCM, it is available in the submicron range, which is advantageous because of shorter diffusion paths, possibly leading to improved electrochemical performance [61]

The high-intensity mixing process is simulated using DEM simulations and stressing conditions are evaluated numerically for the first time. For simulation of submicron-sized particles of LFP, CB, and LIC a reduction in particle number is necessary in order to reduce the simulation time, which was done by applying the coarse-graining method [62–65]. Thus, the particle number can be drastically reduced which greatly shortens the computation time. In addition, a force scaling approach is introduced and validated. Force scaling approaches are necessary when coarse-graining is applied because the forces acting on the coarse grains are different from those acting on the original particles [66]. Force scaling was already applied to DEM contact models, with scalings of normal and tangential contact forces of f^2 [67,68] or f^3 [69–71]. Adhesive forces can be described using the JKR model [72] for which a scaling of f^2 [73] as well as f^3 [74] can be found. For the DEM simulations, a calibration of the mechanical parameters and the static and dynamic properties is necessary to realistically represent the behavior of the particles [75]. For this purpose, shear tester, rotating drums, and cylinder pull-up tests can be used to determine friction parameters such as rolling friction μ_r, mechanical parameters such as elastic modulus E, and adhesion parameters such as surface energy Γ [76–78]. For the calibration of the heteroaggregates, rotary drum tests, cylinder pull-up tests, and compaction tests were performed within this study. The produced heteroaggregate mixtures are built into full cells, cycled, and the influence of mixing time and rotational speed on the electrochemical performance is investigated. The electrochemical data is then matched to simulated stressing of the particles and specific energy data.

With the calibration of static, dynamic, and mechanical behavior of the heteroaggregates, the introduction of the scaling approach, the investigation of stressing mechanisms in the high-intensity mixer, and

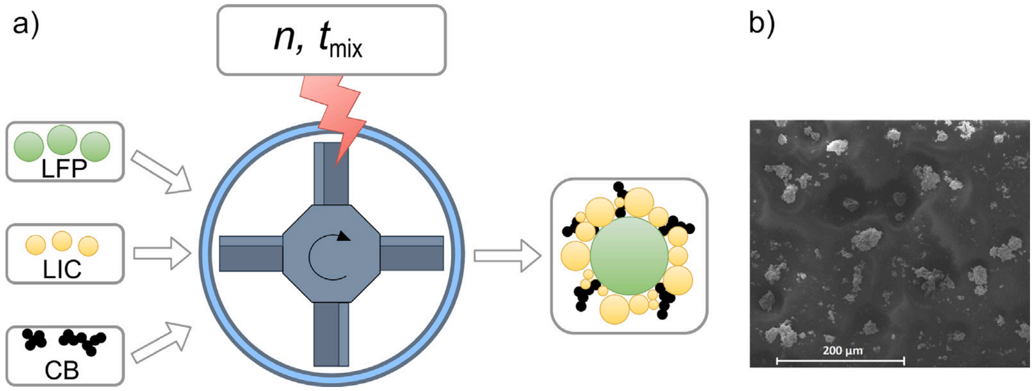

Fig. 1. Production process of heteroaggregates. (a) Schematic representation of the production process via high-intensity mixer. (b) SEM image of the produced heteroaggregates at 10 000 min^{-1}.

the link between simulation and experiment based cycling data, this publication lays the foundation for simulations as well as experimental investigations on the topic of ASSB production via high-intensity mixers. Thus, this is a fundamental step towards understanding and implementation of larger-scale ASSB production which is not established yet [79].

2. Materials and experimental methods

2.1. Preparation of LIC and heteroaggregate mixture

For the preparation of the heteroaggregates, the solid electrolyte LIC was synthesized in-house following the water-mediated synthesis reported by Li and Liang et al. [40]. The starting components LiCl (Fisher Scientific, 99%) and $InCl_3$ (Fisher Scientific, 99.995%) with a mass ratio of 0.365:0.635 (LiCl:$InCl_3$) were mixed in deionized water using stir fish at 40 °C for 2 h. Then, the mixture was predried at 80 °C with the vessel open for 24 h. After predrying, the generated LIC was post-dried for 4 h at 200 °C under vacuum. To produce the desired particle size of 0.1 μm–1 μm, the LIC was comminuted for 4 min at 600 rpm in a planetary ball mill (Pulverisette 7) using 5 mm sized ZrO_2 milling balls before it was further processed and stored under dry room atmosphere.

The synthesized LIC was mixed with LFP (Johnson Matthey Battery Materials GmbH, Carbon–Lithium–Iron–Phosphate, x_{50} = 0.5 μm) and CB (IMERYS, SUPER C65) to prepare heteroaggregates with a mass ratio of 58:4:38 (LFP:CB:LIC). The process for the production of cathode composite heteroaggregates is schematically displayed in Fig. 1a. The Picoline with Nobilta attachment (Hosokawa Alpine AG) was used for the process. It consists of a rotating structure used to mix the components and a horizontally arranged mixing drum that is cooled by a cooling jacket. Cooling is necessary because of the large heat dissipation during intensive mixing as all mechanical energy is dissipated into heat [53]. Influencing process parameters are the rotational speed n and the mixing time t_{mix}.

Heteroaggregates were produced via mixing a total 15 g of powder at rotational speeds n of 1000 min^{-1}–10 000 min^{-1} while mixing for a total of t_{mix}= 60 min. A SEM image of the produced heteroaggregates at 10 000 min^{-1} can be observed in Fig. 1b. The particle size distribution of the heteroaggregates was measured by laser diffraction (Horiba LA-960), which can be taken from the supporting information. From this, a median aggregate size x_{50} of (18.77 ± 0.26) μm is obtained.

2.2. Cell assembly and electrochemical characterization

The full cell assembly was carried out inside an argon-filled glovebox with impurity levels of $p(O_2)/p$ < 1.0 ppm and $p(H_2O)/p$ < 1.0 ppm. First, 35 mg of Li_6PS_5Cl (NEI Corp., NJ, USA) were filled in the press cell and tightened by hand. Then, 35 mg of in-house synthesized LIC were filled over the pressed argyrodite and again tightened by hand. After that, 11 mg of the heteroaggregate mixture were evenly distributed on top of the LIC layer as cathode composite. The whole setup was again uniaxially compacted under a pressure of 380 MPa for 3 min. A In/(InLi) anode was made by placing indium foil (chemPUR, 100 μm thickness, 99.999%) with a diameter of 9 mm, followed by lithium foil (China Energy Lithium, 100 μm thickness, 99.9%) with a diameter of 6 mm on the other side of the Li_6PS_5Cl layer. The cell was fixed in a steel frame with 4 Nm for about 30 minutes to enable diffusion and thus the formation of the indium lithium alloy with a stable potential of 0.62 V vs. Li^+/Li. Before cell cycling the steel frame was tightened with about 25 MPa. The cells were cycled at 0.1 C for 10 cycles between 1.9 V and 3.4 V vs. In/InLi.

2.3. Calibration experiments

For the simulation of the high-intensity mixing process via DEM simulations, calibration of the heteroaggregate powder was done via experiments which were simulated to match the experimental behavior. For this purpose, the dynamic angle of repose (DAOR) was measured by a rotating drum (GranuDrum), the static angle of repose (SAOR) by a cylinder pull-up test, and the mechanical properties by a compaction test.

The DAORs were determined at rotational speeds of 5 min^{-1}, 10 min^{-1}, 15 min^{-1}, and 60 min^{-1}. The drum was filled with the heteroaggregate powder, which showed an experimentally measured bulk density $\rho_{b,exp}$ of (701.83 ± 16.95) kg m^{-3}, and 20 images were taken at each rotational speed and the average of the DAORs were taken from each rotational speed. The determination of the SAOR was performed using an in-house measurement set-up [80]. This involves placing a cylinder on a podest, filling it with the heteroaggregate powder, then pulling it upward at a constant speed, and then determining the SAOR. The mechanical properties of the heteroaggregates were analyzed using a ZWICK material testing machine (ZwickRoell GmbH & Co. KG). For this, the lower die was filled with 1 g of the cathode mixture and then compacted with the top stamp, recording the force–displacement curve.

3. Simulation methods

3.1. Calibration of coarse-grained particles

The calibration of the tests performed for calibration purposes was carried out via DEM simulations, in which a three-step procedure consisting of DAOR via rotating drum, SAOR via cylinder pull-up test, and a compaction test were performed (Fig. 2). The simulations were carried out using the software Rocky version 2023 R1, in which the Hertz–Mindlin and the JKR contact model were applied. A more detailed description of the simulation process and the CAD models used can be found in the supporting information.

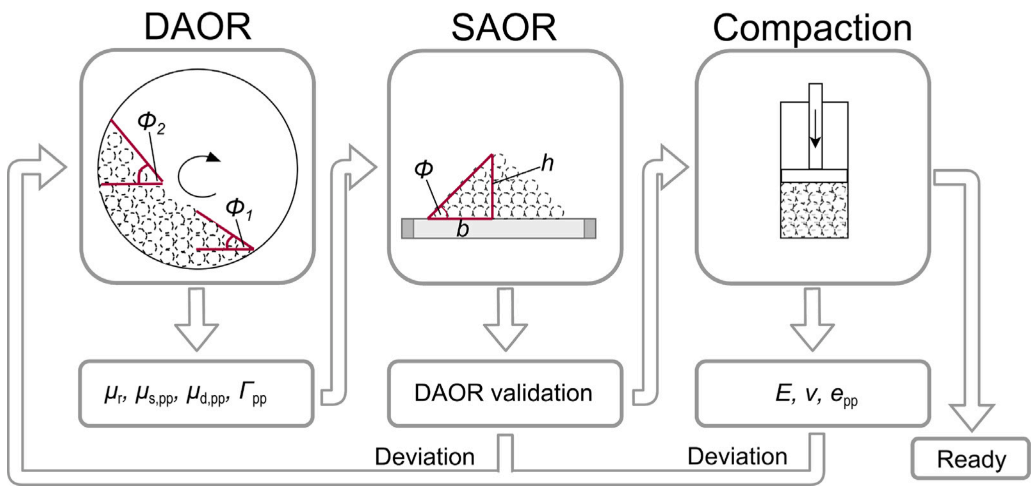

Fig. 2. Calibration procedure for the DEM parameters.

For the DAOR, the filling ratio of simulation and experiment was first checked, and it was found that the measured bulk density of the aggregates of 701.83 kg m^{-3} did not result in the correct filling ratio for the DAOR experiment. This is because the powder itself is compacted by the rotation of the drum, so the bulk density for the simulation was adjusted slightly to $\rho_{b,sim}$ = 750 kg m^{-3}, resulting in the correct filling ratio (supporting information). With a bulk porosity of $\epsilon_{bulk} = 0.4$ this results in a particle density of 1250 kg m^{-3} including pores and an inner porosity of $\epsilon_{inner} = 0.59$. A sensitivity study was performed for the DAOR (supporting information) to determine which parameters have a great influence on the DAOR ϕ_1 and ϕ_2. These parameters which are friction and cohesion parameters like $\mu_{s,pp}$ and Γ_{pp} were used for carrying out a Design of experiment (DoE) in which 16 simulations were conducted. The DoE was created using the pyDOE bibliography from python. For the DAOR simulation a 5 mm wide slice of the actual 20 mm depth rotating drum was simulated and periodic boundaries were set in y-direction. This was done to shorten computation time. The DAOR simulation was performed for 2 s. Particles were first introduced into the system by continuous injection and subsequently the drum was rotated. For calibration, the rotational speed of 10 min^{-1} was chosen. The other rotational speeds served as validation.

After calibrating the DAOR via DoE, the calibrated parameters were used to determine the SAOR via DEM simulations. For the simulation, a mass of 10 g of particles was inserted. The cylinder was lifted at a speed of 0.0075 m s^{-1} and the resulting SAOR was compared to experimental results. If the SAOR of simulation and experiment did not match, parameters were recalibrated. If they matched, the compaction test was simulated and compared with the experimental results. During the simulation of the compaction test, the lower die was filled with 1 g particles and the upper stamp was moved down at a speed of 4.17 mm s^{-1} until the experimentally measured distance between the upper and lower die of ≈6 mm was reached. Then the upper stamp was moved up again. During the process, the force–displacement curve was recorded and compared with the experimentally measured curve for calibration. Through the compaction test, the elastic modulus E, Poisson's ratio ν, and coefficient of restitution e_{pp} were determined. The compaction test was performed last because the mechanical parameters are considered to have less influence on the angles of repose [81].

3.2. DEM set-up of the high-intensity mixer

The CAD model of the high-intensity mixer can be seen in Fig. 3a. It shows the rotating structure with a mixing drum, where the gap between the rotational device and the drum is in the range of 1.532–1.895 mm. The mixer was cut to a fifth of its original size to save computing time and periodic boundary conditions were applied in the y-direction (Fig. 3b). This reduction in mixer size meant that only 3 g of material had to be simulated instead of the 15 g of material used in the experiment.

The simulation was performed for 1 s. 36670 particles were first introduced by continuous injection (0.0 s–0.1 s) and then the mixer was rotated for 0.9 s (0.1 s–1.0 s) at various rotational speeds n ranging from 1000 min^{-1} to 10 000 min^{-1} in increments of 1000.

The size of the particles was chosen to be 0.5 mm, which is about 26.65 times larger than the original heteroaggregate size. Increasing the particle size for computational time reduction is also referred to as coarse-graining [82]. This coarse-grain size was chosen because of a tradeoff between accuracy and computation time (supporting information). If the particles were too large, they would jam in the gap, which would lead to high stress intensities acting on the particles. If they were too small, the simulation time would increase strongly, which is due to two facts. First, the simulation timestep Δt_{sim} is directly related to the particle size (Eq. (1)). In this equation, R^* represents the equivalent radius of the particles, ρ represents the density, G^* represents the equivalent shear modulus, and ν represents the Poisson's ratio. A decrease in the particle size means a decrease in the simulation timestep and thus, an increase in the simulation time [83].

$$\Delta t_{sim} = 0.2 \cdot \frac{\pi \cdot R^* \cdot \sqrt{\frac{\rho}{G^*}}}{0.8766 + 0.1631 \cdot \nu} \tag{1}$$

Second, decreasing particle size means that more particles are required to simulate the same mass of particles leading to a strong increase in simulation time. The sharp exponential increase in simulation time can be observed for both CPU and GPU processors with a particle size smaller than 0.5 μm (supporting information).

The particle shape was chosen to be spherical, although the heteroaggregates have a non-spherical structure. This was done for computational time reasons, as non-spherical particles created in Rocky required much longer computational times (supporting information). To account for the non-sphericity, the rotational resistance was modeled via a rolling friction model [84]. In the present study the linear spring rolling limit was applied and rolling friction was calibrated within calibration experiments and simulations.

3.3. Force-scaling approach

When coarse-graining is applied, the coarse-grain factor f can be defined, which is the quotient of the diameter of the coarse grains d_{CG} (CG = coarse grain) by the diameter of the original particles d_0 (0 = original particle).

$$f = \frac{d_{CG}}{d_0} \tag{2}$$

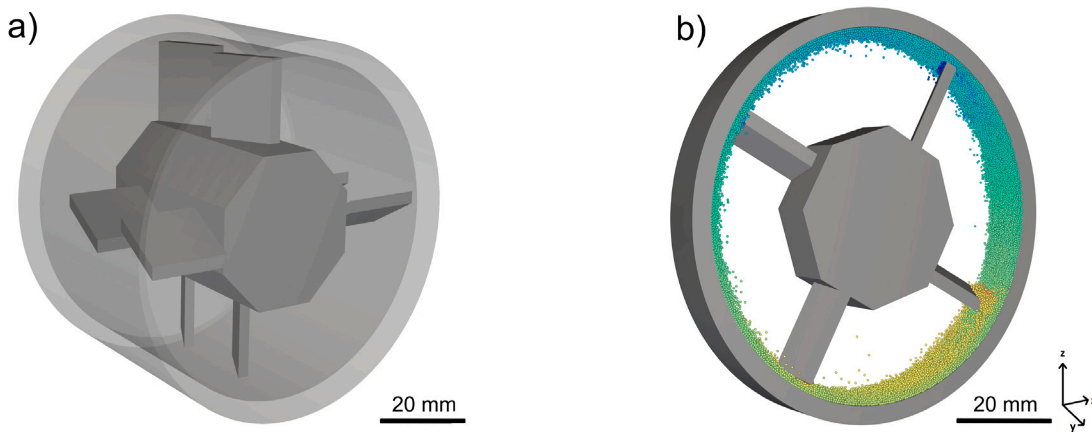

Fig. 3. CAD model of the high-intensity mixer. (a) Complete mixer. (b) Reduced simulation domain.

Due to coarsening the particles by f, the contact forces of the coarse particles differ from those of the original particles [66]. Therefore, a force scaling approach was developed and applied that takes into account that larger particles experience higher forces than smaller particles during collision. The force scaling approach is based on the assumption that the overlap of the coarse grains δ_{CG} scales with the coarse grain factor f, where δ_0 represents the overlap of the original sized particles.

$$\delta_{CG} = \delta_0 \cdot f \tag{3}$$

For particle–particle interaction forces, the assumption is made that, the collision duration τ, scales with f, which goes back to energy conservation [67].

$$\tau_{CG} = \tau_0 \cdot f \tag{4}$$

The equivalent elastic modulus E^* as well as the equivalent shear modulus G^* are considered to be constant. This leads to the Hertz contact force F_n^{Hertz} and the Mindlin contact force $F_t^{Mindlin}$ [85] being scaled with a factor of f^2 as also declared in Bierwisch et al. [68]. The JKR adhesion force F_n^{JKR} includes the surface energy Γ, which scales with f due to dimensional analysis, which also scales the JKR contact force with f^2 as suggested by Washino et al. [73]. The procedure of scaling the contact forces is shown on the basis of the Hertz contact force in Eq. (5).

$$F_{n,CG}^{Hertz} = \frac{4}{3} E^* \cdot \sqrt{R_0^* \cdot f \cdot \delta_{n,0} \cdot f} \cdot \delta_{n,0} \cdot f = F_{n,0}^{Hertz} \cdot f^2 \tag{5}$$

In this, R_0^* refers to the equivalent radius of the original colliding particles, $F_{n,CG}^{Hertz}$ is the normal contact force of the coarse grain calculated via the Hertz contact force model, and $\delta_{n,0}$ represents the overlap of the original particle in the normal direction. The general form of scaling the Hertz contact force, the Mindlin, and the JKR contact force can be derived from Eq. (6). In this F_{CG}^X represents the contact force of the coarse grain and F_0^X stands for the force acting on the original sized particles. For more details on the scaling approach, refer to the supporting information.

$$F_{CG}^X = F_0^X \cdot f^2 \text{ with X = Hertz/Mindlin / JKR} \tag{6}$$

The scaling of the contact forces with f^2 leads to the fact that the stress σ_{CG} acting on the coarse grains is scale-independent, meaning that it is equal to the stress acting on the original particle σ_0 (Eq. (7)). Here, A refers to the cross sectional area of the particle which scales with f^2. The size independent scaling is also true for the specific energy per collision of the coarse grain $E_{m,CG}$. $E_{m,CG}$ is composed of the contact force F_0 scaled by f^2, the overlap δ_0 scaled by f and the mass of the original particle $m_{p,0}$ scaled by f^3 (Eq. (8)).

$$\sigma_{CG} = \frac{F_0 \cdot f^2}{A \cdot f^2} = \sigma_0 \tag{7}$$

$$E_{m,CG} = \frac{\int F_0 \cdot f^2 \cdot d\delta_0 \cdot f}{m_{p,0} \cdot f^3} = E_{m,0} \tag{8}$$

The fact that the stress and the specific energy per collision are scale-independent can be used to evaluate the stresses acting on the real particles in the high-intensity mixer, although coarse grain particles are simulated. This offers the great advantage that significantly fewer particles can be simulated and thus a lot of computational time can be saved.

The scaling approach was validated by colliding two particles with the same size, a defined mass by keeping the density constant at ρ = 1250 kg m^{-3}, a defined velocity of v_{part} = 0.1 m s^{-1}, and a defined distance of 10 mm (supporting information). The force–time curve was plotted to obtain dependencies between the different particle sizes. To avoid any influence from friction, these values were set to small values (0.001). For the Hertz model and the JKR model, the particles were collided in the normal direction and the normal force for the Hertz model and the adhesive force for the JKR model were evaluated. The Mindlin model was validated by evaluating the tangential force of two colliding particles moving towards each other with an offset in y-direction equal to their radius. The evaluation confirmed that the contact forces scale with a factor of f^2, the stress is independent from particle scaling, thus it scales with f^0, the impulse scales with f^3, and collision time scales with f^1 (supporting information).

3.4. Evaluation and export of force and power data

The evaluation of the stresses and specific energies acting on the particles during a collision was performed in the steady state of the mixer. This is reached at the lowest simulated rotational speed of 1000 min^{-1} after about 0.5 s simulation (supporting information). To allow for some tolerance, the data were recorded in the period from 0.7 s to 1 s. During this time, the mean frequency $\overline{c_i}$, the mean collision duration $\overline{\tau}$, force normal F_n, force tangential F_t, and the normal P_i^{normal}, shear P_i^{shear}, and dissipation power $P_i^{dissipation}$ acting on the particles in a time interval of Δt_{out} = 0.005 s were recorded and stored per particle. The power values are the sum of all collision energies W_k^α acting on a particle i during the interval Δt_{out} divided by the interval.

$$P_i^\alpha = \frac{\sum_{k=1}^{N_{k,i}} W_k^\alpha}{\Delta t_{out}} \text{ with } \alpha = \text{normal/shear/dissipation} \tag{9}$$

The mean frequency $\overline{c_i}$ represents the quotient of the collision number $N_{k,i}$ per particle i during the output interval, divided by Δt_{out}. Dividing the sum of all collision durations of particle i during Δt_{out} by the collision number of that particle $N_{k,i}$, results in the mean collision duration $\overline{\tau}$.

$$\overline{c_i} = \frac{N_{k,i}}{\Delta t_{out}} \tag{10}$$

Table 1
Experimental results of the DAOR determination.

Rotational speed [min^{-1}]	Sequence 1		Sequence 2	
	Angle Φ_1 [°]	Angle Φ_2 [°]	Angle Φ_1 [°]	Angle Φ_2 [°]
5	31.20 ± 3.99	54.46 ± 4.92	29.65 ± 4.62	51.87 ± 4.18
10	30.08 ± 5.87	57.38 ± 4.74	30.32 ± 5.29	56.03 ± 5.30
15	29.05 ± 4.44	61.32 ± 4.62	29.14 ± 4.97	62.03 ± 6.16
60	14.78 ± 3.49	65.75 ± 2.10	14.60 ± 3.64	65.14 ± 2.62

$$\overline{\tau} = \frac{\sum_{k=1}^{N_{k,i}} \tau_i}{N_{k,i}} \tag{11}$$

Force normal F_n and force tangential F_t are the sum of all impulses J_i acting on a particle i during Δt_{out}, divided by the output interval.

$$F_{x,i} = \frac{\sum_{k=1}^{N_{k,i}} J_i}{\Delta t_{out}} \text{ with } x = n/t \tag{12}$$

These data were recorded in Rocky, exported, and converted to collision-based specific energies $E_{m,i,collision}$ and collision-based stress values $\sigma_{x,i,collision}$ using an in-house python script. For the specific energies per collision, the power value for each particle is divided by the mean frequency $\overline{c_i}$ and by the mass of the particle m_p. These collision-based specific energies are referred to below as stress intensity SI according to the stressing model of Kwade [86].

$$E^{\alpha}_{m,i,collision} = \frac{\sum_{k=1}^{N_{k,i}} W^{\alpha}_k}{\Delta t_{out}} \cdot \frac{\Delta t_{out}}{N_{k,i}} \cdot \frac{1}{m_p} = \frac{\sum_{k=1}^{N_{k,i}} W^{\alpha}_k}{N_{k,i} \cdot m_p} = SI^{\alpha} \tag{13}$$

The calculation of the collision-based stress values $\sigma_{x,i,collision}$ is done by dividing the force values with $\overline{c_i}$ and $\overline{\tau}$ and by the cross-sectional area A of the particles.

$$\sigma_{x,i,collision} = \frac{\sum_{k=1}^{N_{k,i}} J_i}{\Delta t_{out}} \cdot \frac{\Delta t_{out}}{N_{k,i}} \cdot \frac{N_{k,i}}{\sum_{k=1}^{N_{k,i}} \tau_i} \cdot \frac{1}{A} = \frac{\sum_{k=1}^{N_{k,i}} J_i}{\sum_{k=1}^{N_{k,i}} \tau_i \cdot A} \tag{14}$$

To account for the fact that each particle undergoes a number of $N_{k,i}$ collisions during Δt_{out}, the stress intensities SI and stress values per particle $\sigma_{x,i,collision}$ are stored $N_{k,i}$ times in a new list. This new list is then used to generate the number-based distribution based on a logarithmic distribution of the x-axis. To account for the fact that the data acquisition takes place in a time span of 0.3 s, the distribution is divided by 0.3 to obtain the collision counts per second of mixing. This is referred to below as the stress frequency SF, which indicates how often a given stress intensity SI occurs during a given mixing time [86].

4. Results and discussion

4.1. Calibration of particle behavior

For the calibration of the DAOR, the experimental results of the rotating drum were first evaluated (Table 1). The rotational speed of 10 min^{-1} was chosen for the calibration of the powder. The angles of repose to be set were $\Phi_1 = 30.20°$ and $\Phi_2 = 56.70°$, which are the mean values of sequence 1 and sequence 2. Relatively high standard deviations of up to 5.87° at 10 min^{-1} were determined which is based on the strongly cohesive behavior of the powder, leading to a non-uniform behavior in the rotating drum.

For the calibration of the simulation parameters, a sensitivity study was first performed (supporting information). It was found that the main parameters influencing the DAOR are the rolling friction μ_r, the static and dynamic friction between the particles $\mu_{s,pp}$, $\mu_{d,pp}$, as well as the surface energy Γ_{pp} between the particles. Other parameters which are mostly particle–wall interactions like the static and dynamic friction between particles and the wall $\mu_{s,pw}$, $\mu_{d,pw}$ play a minor role and were set constant. By varying the particle–particle parameters a DoE was conducted, in which the rolling friction in a range of μ_r= [0.1 – 0.5], the static friction between the particles in a range of $\mu_{s,pp}$= [0.1 – 0.9], the dynamic friction between the particles in a range of $\mu_{d,pp}$= [0.1 – 0.9], and the surface energy between the particles Γ_{pp} = [0.0005 – 0.005] J m^{-2} were varied. In addition, the condition that static friction is greater than or equal to dynamic friction was applied. This way it was possible to determine the parameter combination that can correctly represent the behavior of the powder at all rotational speeds ranging from 5 min^{-1} to 60 min^{-1} (Fig. 4 and Table 2). The calibrated parameters are the rolling friction $\mu_r = 0.2$, the static friction between particles $\mu_{s,pp}$= 0.47, the dynamic friction between particles $\mu_{d,pp}$= 0.44, and the surface energy between particles $\Gamma_{pp} = 0.0045$ J m^{-2}. In Fig. 4 the mean values of each rotational speed are drawn.

Table 2
Simulation results of the DAOR determination.

Rotational speed [min^{-1}]	Angle Φ_1 [°]	Angle Φ_2 [°]
5	30.82 ± 4.11	53.77 ± 1.64
10	30.35 ± 2.95	57.22 ± 3.18
15	29.26 ± 2.36	61.50 ± 2.72
60	15.50 ± 5.33	65.61 ± 3.17

Within the simulation of the DAOR, a strongly inhomogeneous behavior of the particles over time was observed. This coincides with the experimental observations in which high standard deviations were calculated. The inhomogeneity mainly comes from applying the surface energy Γ_{pp} which reinforces the need for using adhesive models like the JKR model. A similar conclusion was reached by Hoshishima et al., who also used the JKR model and calibrated strongly cohesive particles by adjusting the surface energy and the rolling friction [78].

The calibrated parameters were used for being validated by simulating the SAOR and comparing the results with the experimental determined SAOR. Fig. 5a shows that the experimentally determined SAOR (55.58° ± 1.44°) can be reproduced by the simulation (54.21 ± 1.14)° confirming that the parameters were correctly calibrated.

For calibrating elastic modulus E, coefficient of restitution e_{pp}, and Poisson's ratio ν, the powder compression was considered. The simulation of the force–displacement curve and the comparison with the experimentally recorded curve show that the mechanical parameters were calibrated correctly (Fig. 5b). In this context, the elastic modulus E = 25 MPa, the coefficient of restitution between particles e_{pp} = 0.3, and the Poisson's ratio ν = 0.3 were calibrated. An overview of all parameters can be found in Table 3. The deviation in the force between experimental and simulated compaction at the beginning of the force–displacement can be attributed to the use of coarse-grain particles. In reality, a highly porous particle network (ϵ_{total} = 0.75) with different particle sizes is present, which is initially rearranged during compaction. This effect is difficult to be reproduced with the monomodal coarse-grain particles, which explains why the increase in force occurs earlier. At higher compaction, however, the simulated and the experimental force curve coincide, which is why the mechanical properties of the original particles can be seen as sufficiently calibrated.

The relatively low elastic modulus of 25 MPa is due to the fact that the heteroaggregate powder represents a highly porous bulk which, when compacted, initially offers hardly any counterforce for compaction. Only after the particles have been rearranged and the adhesive forces have been overcome, a significant increase in force takes place, which is characterized by the stronger material bridges within the aggregates and compaction of the already densified bulk. This means that

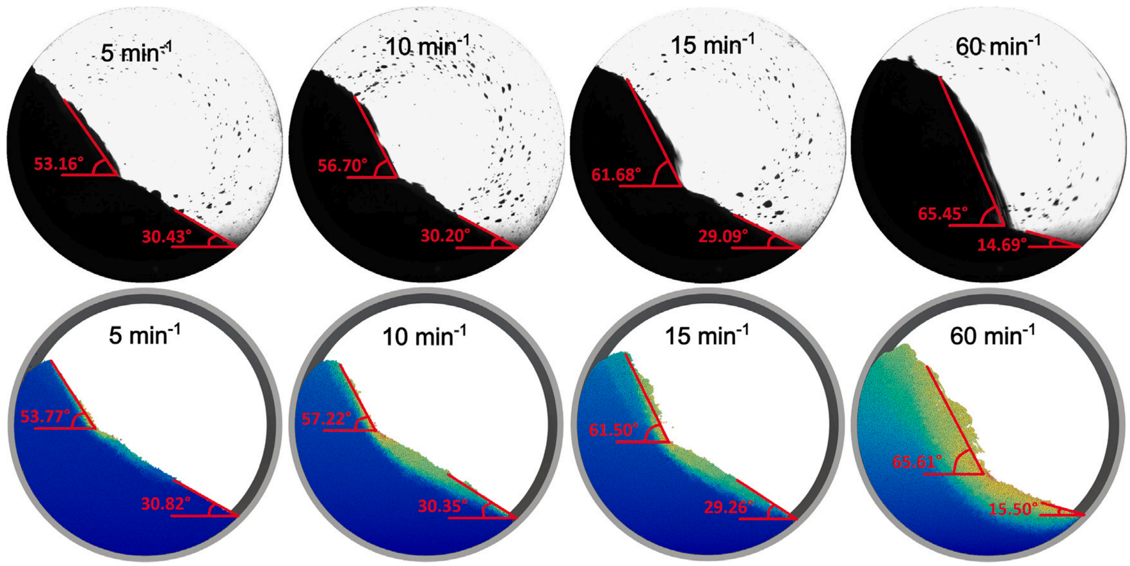

Fig. 4. Experimental and simulational determination of the DAORs.

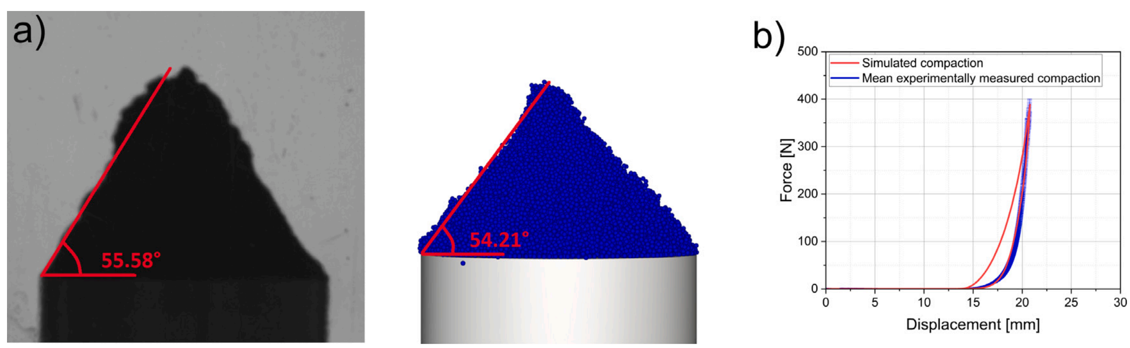

Fig. 5. Calibration of DEM parameters. (a) Determination of SAOR via cylinder pull-up test. (b) Calibration of compaction behavior via force–displacement curve.

Table 3
Calibrated parameters and material properties of the coarse-grains.

	Mixer wall	Coarse-grain
Pure density ρ_0 [kg m^{-3}]	4510	3047
Particle density ρ_p [kg m^{-3}]	–	1250
Bulk density $\rho_{b,sim}$ [kg m^{-3}]	–	750
Elastic modulus E [MPa]	120 000	25
Poisson's ratio ν [–]	0.3	0.3
	particle–wall ($_{pw}$)	particle–particle ($_{pp}$)
Restitution coefficient e [–]	0.3	0.3
Rolling friction μ_r [–]	–	0.2
Static friction μ_s [–]	0.5	0.47
Dynamic friction μ_d [–]	0.3	0.44
Surface energy Γ [J m^{-2}]	0.001	0.0045

the elastic modulus is not determined for the pure material behavior, but for the behavior of the heteroaggregate powder under compaction and stress.

4.2. Stressing conditions in the high-intensity mixer

The calibrated particles were used for simulating the high-intensity mixing process and the stress intensities SI acting on the particles during 1 s of mixing in the whole simulated domain were determined. From the consideration of the normal stress intensity (SI^{normal}) distribution (Fig. 6a) and shear stress intensity SI^{shear} distribution (Fig. 6b), it can be seen that the SI increases with increasing rotational speed n. Moreover, the stress frequency SF also increases with increasing rotational speed. Both increase as the particles are accelerated more strongly and thus collide more frequently and more strongly. This is mainly due to the increasing relative velocity v_{rel} between the rotor and the particles, which leads to higher stress intensities acting on the particles. The high power input also leads to a highly energy-intensive energy cascade, which is propagated through particle–particle collisions. This energy cascade, as well as dissipation of energy through friction, inelastic collisions and adhesion, is shown as dissipation stress intensity ($SI^{dissipation}$) distribution (supporting information).

The stress intensity distributions show a non-monomodality especially at low rotational speeds. This indicates that two stress mechanisms occur, which might be compression and shearing of the particles in the gap and the impact of the particles at the rotor. According to Rumpf, stressing of the particles occurs by compression and shear between two solid surfaces or by impact at one solid surface [87]. At higher rotational speeds, both stress mechanisms overlap more strongly, which is why the non-monomodality of the SI distribution can only be recognized to a limited extent. Nevertheless, both stress mechanisms also occur here, which is because for the evaluation of the stress intensities, mean values were taken over the output interval Δt_{out}. This means that the distribution in reality would be even broader and the non-monomodality would be more visible.

For validation of the two stress mechanisms, the mean stress intensities as well as the mean translational velocity v_{part} are evaluated at specific areas during the mixing process (n = 10 000 min^{-1}), which are Region 1: In gap (black particles; SI_1), Region 2: At rotor blade (white particles; SI_2), Region 3: Outside of gap (blue particles; SI_3) (Fig. 7).

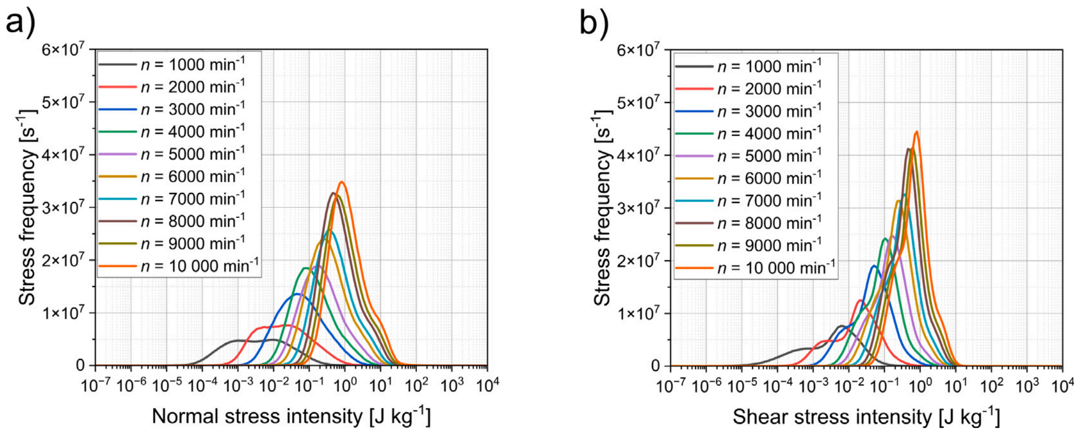

Fig. 6. Stressing conditions in the entire high-intensity mixer. (a) Normal stress intensity distribution at different rotational speeds n. (b) Shear stress intensity distribution at different rotational speeds n.

Table 4
Mean stress intensities and particle velocity at different mixer regions.

	Region 1	Region 2	Region 3
Normal stress intensity SI_x^{normal} [J kg^{-1}]	7.410	49.430	0.236
Shear stress intensity SI_x^{shear} [J kg^{-1}]	6.005	13.893	0.327
Dissipation stress intensity $SI_x^{dissipation}$ [J kg^{-1}]	14.626	51.795	0.764
Translational velocity $v_{part,x}$ [m s^{-1}]	12.134	35.007	9.980

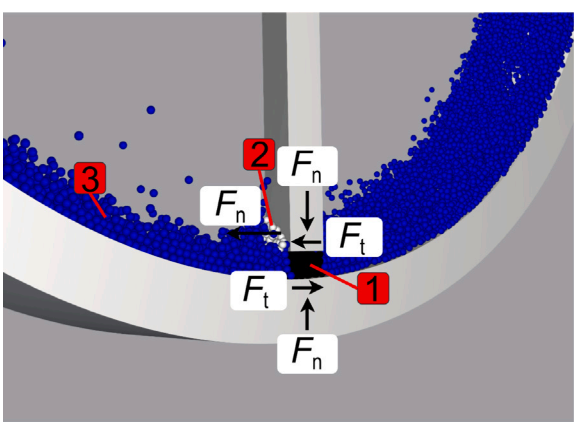

Fig. 7. Stressing conditions in specific areas in the high-intensity mixer.

For more accurate values, the data acquisition interval was reduced to Δt_{out} = 5 μs and data were evaluated over a period of 0.3 s of mixing (0.7 s to 1.0 s).

Stress intensities: The normal stress intensity acting on a particle in the gap (SI_1^{normal} = 7.410 J kg^{-1}) is higher than SI^{normal} outside of gap (SI_3^{normal} = 0.236 J kg^{-1}) and lower than at the rotor blade (SI_2^{normal} = 49.430 J kg^{-1}), (Table 4). The difference between stress intensity in the gap (SI_1^{normal}) and at rotor blade (SI_2^{normal}) confirms that two stress mechanisms take place being high impact stresses at the rotor blade and compression of particles in the gap. In the gap, especially compression acts in the normal direction, which leads to SI_1^{normal} being higher than outside the gap. Outside the gap comparatively low SI^{normal} are present, which only occur due to the particle–particle collisions leading to an energy cascade after high centripetal forces are caused by the rotor. The comparatively high normal stress intensities at the rotor, are caused by the high relative velocity between the rotor tip and the particles.

The mean shear stress intensity is highest at the rotor blade (SI_2^{shear} = 13.893 J kg^{-1}) and lowest outside of the gap (SI_3^{shear} = 0.327 J kg^{-1}). In Region 1, inside the gap, shear stress intensity (SI_1^{shear} = 6.005 J kg^{-1}) is about as high as SI_1^{normal} inside the gap. This is different from Region 2 where SI_2^{normal} is 3.558 times higher than the specific shear energy.

SI^{shear} is strongly dependent on the rotational speed n, which induces a certain shear rate in the gap of the high intensity mixer. The shear rate can be calculated via the rotational speed n, the Radius of the rotor blade R_r, and the gap size h which has a minimum of 1.532 mm.

$$\gamma = \frac{dv}{dh} = \frac{R_r \cdot 2\pi \cdot n}{h} \tag{15}$$

Thus, the shear rate in the gap is in a range of 2 604 s^{-1} and 26 042 s^{-1}. In the literature, particularly high shear rates in high-intensity mixers are attributed to a good dispersing effect and the possibility of aggregating materials [53]. This would make particular sense with respect to the coating of active materials such as NMC by protective coatings such as $LiNbO_3$ [35]. In this case of LFP and LIC, coating/aggregating LFP with LIC or CB can lead to good interfacial contact and thus improved performance of the cathode of an ASSB.

The dissipation stress intensity in the gap ($SI_1^{dissipation}$ = 14.626 J kg^{-1}) is lower than at the rotor ($SI_2^{dissipation}$ = 51.795 J kg^{-1}) and higher than outside the gap ($SI_3^{dissipation}$ = 0.764 J kg^{-1}). High $SI^{dissipation}$ can lead to aggregation processes, whereby aggregates consisting of LFP, LIC, and CB could form [55]. This is because heat dissipation could lead to melting of the particles and thus to sinter-granulation or mechanofusion [53]. This type of aggregate formation is supported by the theory of Thiessen, which explains particle fusion due to internal and external friction and energy dissipation by inelastic collisions between particles [52]. Depending on how the energy is dissipated, comminution processes in which energy is dissipated when particles are broken up are also conceivable. This is more likely at the rotor blade as particles are hit by the fast rotating mixing tool and high SI^{normal} occur of which a lot of energy is dissipated.

Velocity: The translational velocity of the particles v_{part} is highest at the rotor ($v_{part,2}$ = 35.007 m s^{-1}), which is due to the fact that the tip speed of the rotor ($v = 2\pi \cdot n \cdot R_r$ = 39.9 m s^{-1}) is imposed on the particles. With an average of 35.007 m s^{-1}, the velocity of the particles is lower than the rotor tip speed, which is due to the fact that the average velocity is taken over all particles in the considered area in front of the rotor, of which not all of the particles are directly at the rotor tip. The velocity of the particles in the gap ($v_{\mathrm{part},1}$ = 12.134 $\mathrm{m\,s^{-1}}$) is below the average velocity at the rotor, which is due to stronger friction and compression. Outside the gap ($v_{\mathrm{part},3}$ = 9.980 $\mathrm{m\,s^{-1}}$), the lowest average particle velocities occur. This is due to the energy cascade taking place, whereby energy is dissipated through each particle–particle collision and through friction with the wall.

4.3. Applying the stressing model of Kwade to the high-intensity mixing process

The product-related stressing model of Kwade is used and applied to the high-intensity mixing process [88]. Kwade's model states that the specific energy E_m acting on the product mass m_total during a comminution process is the product of the stress number SN and the average stress energy $\overline{SE}$, divided by the product mass m_total [86,88].

$$E_\mathrm{m} = \frac{\overline{SE} \cdot SN}{m_\mathrm{total}} \tag{16}$$

Since in the present work the SE of each collision is already divided by the mass of the particle m_p ($SI = \frac{SE}{m_\mathrm{p}} = E^{\alpha}_{\mathrm{m,i,collision}}$), for the calculation of the specific energy E_m related to the total mass, the particle mass is factored out from the $\overline{SI}$ (supporting information).

From a first physical approach it could be assumed that SN is proportional to n and $\overline{SI}$ in case of impact at rotor tool is proportional to n^2, which is based on kinetic energy approach, and in case of compression and shear is proportional to $n^{2\ldots3}$. For verification or refutation, quantities of the high intensity mixing process were equated with quantities of the stressing model. For this the stress number SN of all particles was calculated for 1 s of mixing for each rotational speed ($SN = \sum SF \cdot t$) and the mean normal stress intensity $\overline{SI}^{\mathrm{normal}}$/shear stress intensity $\overline{SI}^{\mathrm{shear}}$/dissipation stress intensity $\overline{SI}^{\mathrm{dissipation}}$ and the specific normal energy $E^{\mathrm{normal}}_\mathrm{m}$/shear energy $E^{\mathrm{shear}}_\mathrm{m}$/dissipation energy $E^{\mathrm{dissipation}}_\mathrm{m}$ were calculated from the $SF - SI$ distributions from Section 4.2 and plotted over the rotational speed n (supporting information). By means of an allometric fit, dependencies between the $\overline{SI}, SN, E_\mathrm{m}$ and the rotational speed n could be established.

The stress number SN shows a proportionality of $SN \propto n^{0.62}$ to the rotational speed n. The dependencies for the average normal, shear, and dissipation stress intensity, were derived as follows.

$$\overline{SI}^{\mathrm{normal}} \propto n^{2.17} \tag{17}$$

$$\overline{SI}^{\mathrm{shear}} \propto n^{2.09} \tag{18}$$

$$\overline{SI}^{\mathrm{dissipation}} \propto n^{2.11} \tag{19}$$

Thus, the average stress intensity $\overline{SI}$ increases more strongly with increasing rotational speed n compared to the SN. The dependence of the $\overline{SI}$ on the rotational speed n reveals that the stress intensities are approximately $\overline{SI} \propto n^2$ proportional to the rotational speed. This indicates that the $\overline{SI}$ acting on the particles is proportional to the kinetic energy transferred to the particles by the mixer. Thus, the rotational speed n is proportional to v_rel, which is proportional to the particle velocity v_part ($n \propto v_\mathrm{rel} \propto v_\mathrm{part}$) when the particles hit a static wall/static particle after colliding with the fast rotating mixing tool or another particle. This behavior is otherwise known from stirred media mills, where the stress intensity $\overline{SI}$ is proportional to the stirrer tip speed, which can be seen proportional to the rotational speed n: $\overline{SI} \propto v_\mathrm{t}^2 \propto n^2$ [89].

The product of the dependences of the stress number SN and the stress intensity $\overline{SI}$ on the rotational speed n, yields the dependence of the specific energy E_m on the rotational speed (supporting information).

$$E^{\mathrm{normal}}_\mathrm{m} \propto n^{2.17} \cdot n^{0.62} = n^{2.79} \tag{20}$$

$$E^{\mathrm{shear}}_\mathrm{m} \propto n^{2.09} \cdot n^{0.62} = n^{2.71} \tag{21}$$

$$E^{\mathrm{dissipation}}_\mathrm{m} \propto n^{2.11} \cdot n^{0.62} = n^{2.73} \tag{22}$$

The specific energy acting on the product mass scales with a factor of about $E_\mathrm{m} \propto n^{2.7\ldots2.8}$. Specific energy E_m and specific Power P_m are proportional to each other or can be equated in this particular case, since a time frame of one second is considered. Comparing the simulatively determined E_m with the experimentally determined P_m, a good correlation between experimentally and simulatively determined specific power and its dependence on the rotational speed is obtained, which is that P_m is proportional to $P_\mathrm{m} \propto n^{2.82}$ (supporting information). This is in good correlation with the simulatively determined dependency of specific energy being $E_\mathrm{m} \propto n^{2.7\ldots2.8}$.

In the literature, mainly vertical mixers with high shear have been studied so far, often with lower rotational speeds and larger particles like glass beads with 2 mm bead size. In this context, dependencies of power and rotational speed of $P \propto n^{1.31}$ ($n = 10\ \mathrm{min^{-1}}$–$200\ \mathrm{min^{-1}}$) [90], $P \propto n^{1.65}$ (n = 450 $\mathrm{min^{-1}}$ – 1500 $\mathrm{min^{-1}}$) [91] and $P \propto n^{1.9}$ (n = 100 $\mathrm{min^{-1}}$–600 $\mathrm{min^{-1}}$) [92] were determined. Detailed studies on high-intensity horizontal mixers with much higher rotational speeds and fine cohesive powders such as LFP, LIC, CB are not yet known and may provide further insight into the stresses and stress mechanisms in the future.

4.4. Link between simulated stressing conditions and electrochemical performance of full cells

The formation of aggregates could be caused by aggregation processes or by high adhesive forces between LFP, CB, and LIC. Agglomeration increase with decreasing particle size and since in the present case the particles are mostly in the submicron or lower micron range, high adhesive forces could prevail and lead to aggregate formation [93]. Presumably, both mechanisms occur, which are agglomeration due to high adhesive forces and aggregation due to high collision energies.

Agglomeration, aggregation, comminution, and mixing can lead to improved interfacial contact between LFP, LIC, and CB, resulting in better cell performance. Though, the highly cohesive LIC can form larger, loosely attached agglomerates that can hinder proper ion conduction through the ASSB cathode leading to decreased cell performances. This can be clearly observed for the heteroaggregate cathodes produced at a rotational speed of 1000 $\mathrm{min^{-1}}$ (Fig. 8a,b). The LIC particles form large agglomerates with a size of several tens of μm (Fig. 8a) and these agglomerates cluster in regions extending over several hundreds of μm (Fig. 8b). Also, agglomerated CB is visible which is an indicator for insufficient mixing. Many active material particles remain ionically unconnected and thus cannot take place in the charge/discharge process which should lead to a low cell capacity.

This agglomeration should be largely prevented when high $\overline{SI}$ are applied by high rotational speeds of the mixer and can also induce aggregation between LFP, CB, and LIC particles and a better mixing. This can be observed for the composites prepared at 10 000 $\mathrm{min^{-1}}$ (Fig. 8c,d). The particles appear to be relatively homogeneously mixed, which in turn leads to more heterocontacts between the three components. Thus the incorporation of the LFP particles into the ionic network and thus the cell capacity should be improved.

Furthermore, the different rotational speeds seem to have an impact on the mechanical stability of the composites. For low rotational speeds, the surface of the composites shows cracks and unevenness (Fig. 8b) which is not the case for the composite prepared at 10 000 $\mathrm{min^{-1}}$ (Fig. 8d).

The influence of rotational speed n and mixing time t_mix on cell performance was investigated by cycling of full cells which where built with the powders produced at different rotational speeds n between 1000 $\mathrm{min^{-1}}$ and 10 000 $\mathrm{min^{-1}}$. From the average discharge capacity

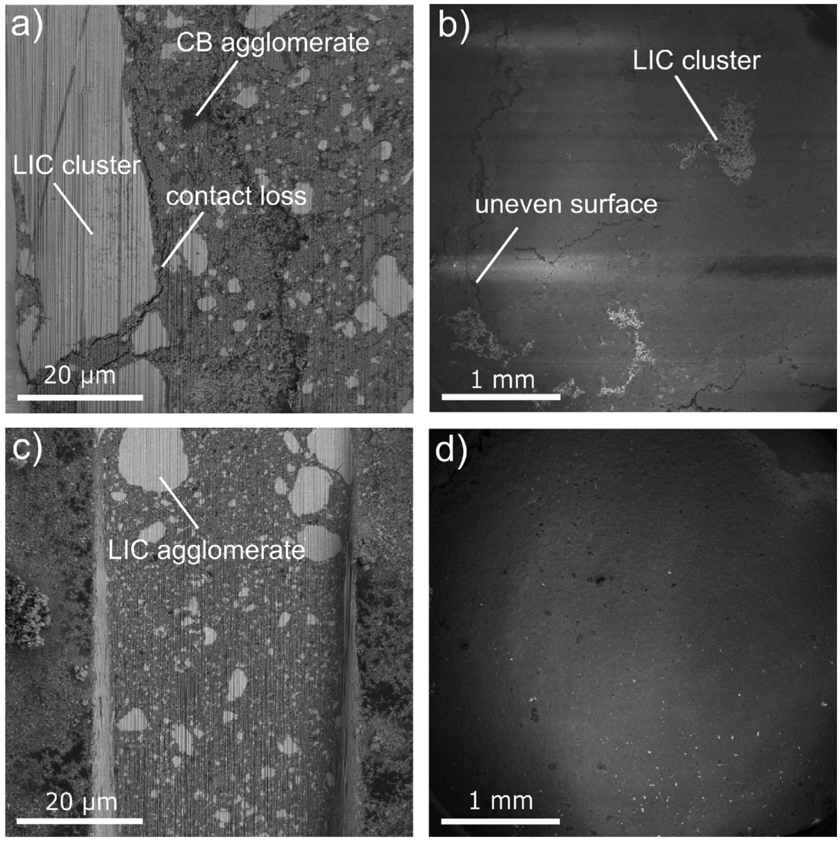

Fig. 8. FIB images of cathodes fabricated at different rotational speeds. (a) Cross-section 1000 min^{-1}. (b) Top view 1000 min^{-1}. (c) Cross-section 10 000 min^{-1}. (d) Top view 10 000 min^{-1}.

$C_{discharge}$ (Fig. 9a), it can be seen that the capacity increases almost linearly with increasing rotational speed ($C_{discharge} \propto n$) and the highest cell capacity is obtained with the cathode mixtures prepared at 10 000 min^{-1} ($C_{discharge}$ = (109 ± 5) mA h g^{-1}). This confirms the microstructural observations (Fig. 8) and the hypothesis that due to the higher rotational speed, better mixing occurs and due to the higher energy input, the probability of aggregation of the materials is increased, both leading to increased interfacial contacts between LFP, CB, and LIC. The improved interfacial contact is also strongly caused by the comminution of the LIC agglomerate clusters, which is evident from the improved cell performance with increasing rotational speed.

The fact that the typical capacity for LFP of 140 mA h g^{-1} [4, 94], already achieved with polymer-based solid electrolytes, cannot be reached in this process even at the highest rotational speed is most likely related to the insufficient size reduction of the LIC particles and thus an insufficient ionic network. As seen in Fig. 8c, larger LIC agglomerates still exist which hinder an optimal ionic connection of the submicron LFP particles. To improve this, the comminution of the solid electrolyte after synthesis needs to be optimized. To this extent other processes than ball milling might need to be explored.

The cell capacity of the cathode mixture produced at 10 000 min^{-1} as a function of the mixing time (Fig. 9b) shows that the discharge capacity of the cathode mixtures increases with increasing mixing time, with a steady state occurring from minute 30 onwards. The mixing time also has an influence on the mixing and aggregation of the materials and thus on the interfacial contact between LFP, CB, and LIC. The longer the mixing time, the better the mixing and the more frequent the collisions between the materials, which can lead to aggregate formation and LIC comminution. This has a positive effect on the performance of the cell, which is reflected in the increased capacity of the full cell.

The reason for the fact that the discharge capacity does not increase any further from minute 30 is that the material increasingly accumulates on the outer wall. Although this results in a continuous compression of the material on the outer wall of the mixer, further mixing and thus aggregation and comminution processes only take place to a minor extent.

The discharge capacities of the cells produced at 5000 min^{-1} indicate an increasing capacity as mixing time progresses and an optimum after t_{mix} = 30 min can be seen. As for 10 000 min^{-1}, the reason is the better mixing and aggregation of the LFP, LIC, and CB materials with mixing time and further comminution of large LIC agglomerates. The reason for the decrease in $C_{discharge}$ after t_{mix} = 60 min could be that at lower rotational speeds, macro-level mixing is not fully present. This would cause the cells built from the mixture after t_{mix} = 60 min to show different performances. This hypothesis is supported by the fact that the standard deviation of the cell after t_{mix} = 30 min and t_{mix} = 60 min almost overlap, which means that the mix after 30 min could give comparable performance to the mix after 60 min. After t_{mix} = 60 min, undesirable granulation effects could also cause the decrease in cell capacity. This suggests that the optimum mixing time for cathode mixing is around t_{mix} = 30 min.

Comparing the influence of mixing time t_{mix} and rotational speed n, it is evident that the discharge capacity of the cells prepared with powder mixed at 10 000 min^{-1} after t_{mix} = 5 min is higher than all other samples produced with powders prepared at 1000 min^{-1} to 7500 min^{-1} after t_{mix} = 60 min. This indicates that the rotational speed, and thus the $\overline{SI}$, is a predominant contributor to the heteroaggregate formation in such a way that good interfacial contact is formed between the materials. This is proved by Fig. 9c,d, which shows the dependence of $C_{discharge}$ on $\overline{SI}$ and SN. For the same SN, the $C_{discharge}$ is lower when a lower $\overline{SI}$ of 0.78 J kg^{-1} is applied (5000 min^{-1}), compared to when a higher $\overline{SI}$ of 3.35 J kg^{-1} is induced by the mixer (10 000 min^{-1}). Considering the dependence of $C_{discharge}$ on the specific energy for different rotational speeds (Fig. 9e, t_{mix} = 60 min)

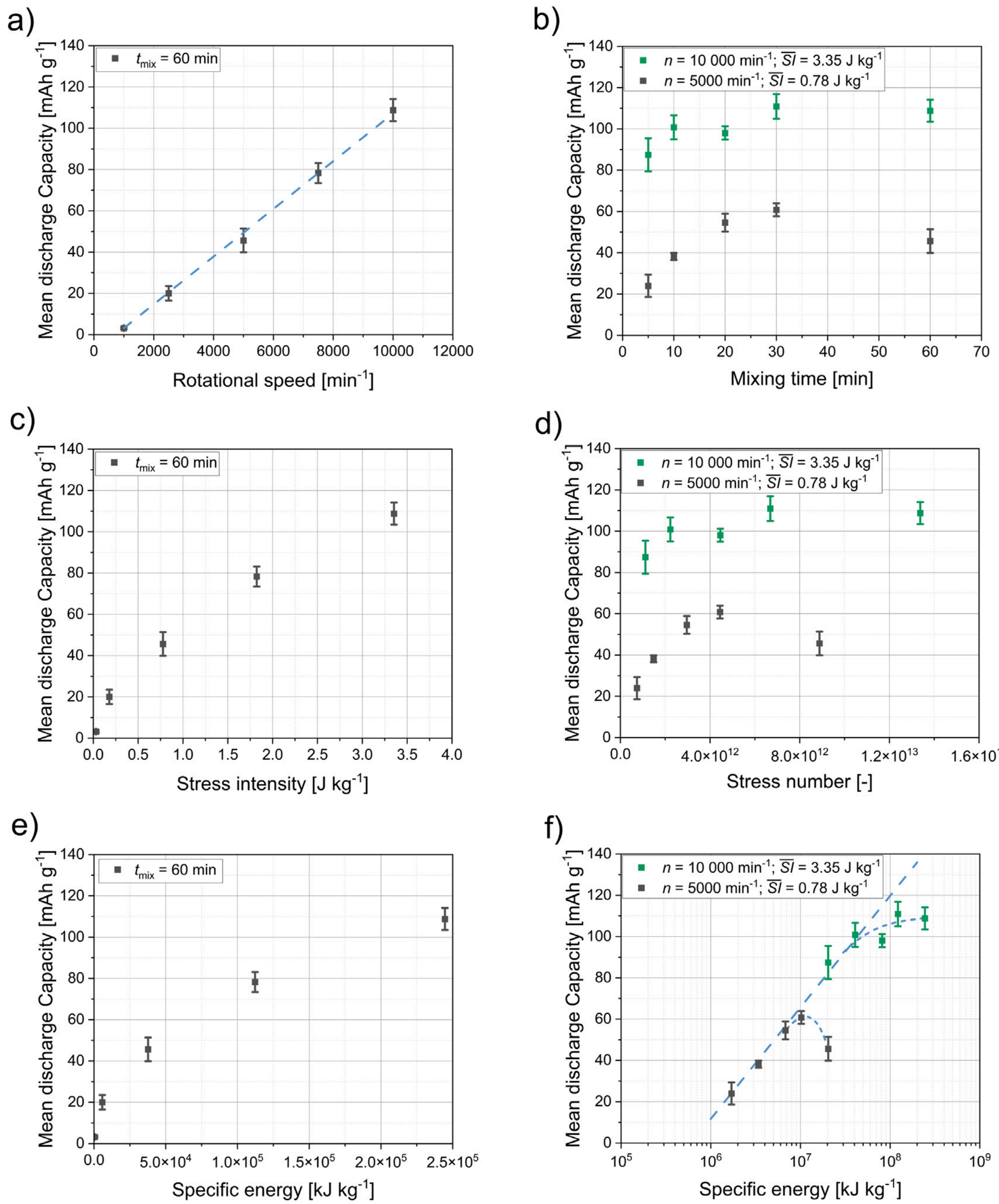

Fig. 9. Mean discharge capacity of heteroaggregate powders. (a) Dependency of discharge capacity on rotational speed (t_{mix} = 60 min). (b) Dependency of discharge capacity on mixing time (t_{mix} = 5 min–60 min). (c) Dependency of discharge capacity on mean stress intensity per particle (t_{mix} = 60 min). (d) Dependency of discharge capacity on stress number (t_{mix} = 5 min–60 min). (e) Dependency of discharge capacity on specific energy (t_{mix} = 60 min). (f) Dependency of discharge capacity on specific energy (t_{mix} = 5 *min*–60 min).

further proves this finding. In other words, a maximum dispersing effect and thus homogeneity of the heteroaggregate structure depends on the applied stress intensity and is achieved after a sufficient stress number or specific energy, respectively. At a chosen stress intensity, even at very high stress numbers no further improvement in dispersing grade and homogeneity of the heteroaggregate structure is possible, since the stress intensity is not able to disperse the very fine primary particles further and particles accumulate on the mixer chamber wall. A similar behavior is known for wet dispersing processes [95]. Moreover, even degradation effects could occur due to unwanted re-agglomeration or granulation (see Fig. 9b, $C_{discharge}$ after t_{mix} = 60 min for 5000 min^{-1}).

However, as long as the maximum degree of dispersion for a given stress intensity is not reached, the specific energy input for a given degree of dispersion as well as the heteroaggregate homogeneity and thus the specific capacity seem to be almost independent of the stress intensity (Fig. 9f). This means that up to this maximum degree of dispersion for a given stress intensity, the effects of increasing the stress number and increasing the stress intensity appear to be comparable. Therefore, for example, doubling the number of stress events has a similar effect as doubling the stress intensity. One possible explanation could be that doubling the $\overline{SI}$ disconnects twice the particle contacts or bonds. In future studies this behavior will further be investigated, which was also found for mechanochemical reactions [58,96].

5. Conclusions

In this study, the high-intensity mixing process for the formation of heteroaggregates consisting of submicron-sized LFP, LIC, and CB was investigated using DEM simulations and mixing experiments. The heteroaggregates are studied for use as cathode material for ASSB. In order to establish a good contact between the active material and the solid electrolyte, a high intensity mixing process is used.

To simulate the mixing process, the heteroaggregates were calibrated in a three-step calibration procedure consisting of the determination of DAOR, SAOR, and a compaction test. Calibrated parameters were the surface energy between particles Γ_{pp}, static/dynamic friction between particles $\mu_{s,pp}$ and $\mu_{d,pp}$, rolling friction μ_r, and mechanical parameters such as elastic modulus E, Poisson's ratio ν, and coefficient of restitution e_{pp}.

The simulation time was reduced by decreasing the simulation domain and by coarse-graining the particles to 26.65 times its original size. The mixing process was simulated at different rotational speeds n, and the resulting stress intensities SI acting on the particles were evaluated. With increasing rotational speed, an increase in stress frequency SF and thus stress number SN as well as in $\overline{SI}$ was observed for $\overline{SI}^{\text{normal}}$, $\overline{SI}^{\text{shear}}$, and $\overline{SI}^{\text{dissipation}}$. It was found that mainly two stress mechanisms occur which are the compression and shearing of particles in the gap between rotor tip and mixing drum and the impact stress outside the gap.

Kwade's stressing model was applied to the mixing process, and dependencies were derived between the rotational speed n and $\overline{SI}$, SN and E_m, showing that $\overline{SI}$ increases more strongly with increasing rotational speed than the SN ($\overline{SI} \propto n^{\approx 2}$, $SN \propto n^{0.62}$). The specific energy scales with approximately $E_m \propto n^{2.7} - n^{2.8}$ which is in good correlation with the experimentally obtained dependency of the specific power on rotational speed ($P_m \propto n^{2.82}$).

The simulated results were matched with experimentally measured electrochemical data, showing that a higher stress intensity and an increasing SN lead to an increased capacity of the formed full cells, with $\overline{SI}$ having a greater influence. This is probably due to comminution processes of larger LIC agglomerates and due to aggregation of LFP, CB, and LIC particles which results from high $\overline{SI}$ leading to aggregation mechanisms through dissipation of energy and adhesive forces. The applied $\overline{SI}$ at 10 000 min^{-1} provides the maximum achievable homogeneity of the mixture and thus the maximum achievable capacity of about 109 mAh g^{-1}. Further increase in capacity may be possible with other mixing devices with a higher $\overline{SI}$, which will be investigated in the future.

With the calibration of the coarse-grains, the implementation and validation of the force-scaling approach, and simulation results on stress mechanisms in the high-intensity mixer, this publication lays the foundation for further investigations in the field of ASSB processing with high-intensity mixers. In the future, the macrosimulation is transferred to the micro level, which could simulate the aggregation/agglomeration of particles. Further experimental investigations will be carried out aiming to increase the electrochemical performance of the full cells with the LFP, LIC, CB system.

CRediT authorship contribution statement

Finn Frankenberg: Conceptualization, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing, Data curation, Validation, Visualization. **Maximilian Kissel:** Conceptualization, Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Christine Friederike Burmeister:** Conceptualization, Methodology, Supervision. **Mark Lippke:** Supervision. **Jürgen Janek:** Conceptualization, Project administration, Resources, Supervision, Funding acquisition, Writing – review & editing. **Arno Kwade:** Conceptualization, Project administration, Resources, Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Finn Frankenberg reports financial support was provided by Deutsche Forschungsgemeinschaft (DFG). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.powtec.2024.119403.

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